

COHERENCE ATLAS

A Cosmological Field Computer

A founding document, research map, and open invitation
to scholars, practitioners, and builders at the confluence
of Vedic knowledge and living technology.

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Gainesville, Florida — kanjira — Dvādaśī śukla, Uttara Bhādrapadā

Invocation

The satisfaction of the divine couple does not lie in union alone, nor in separation alone, but precisely in the living cycle of their interchange — a fully complete energetic movement in which union and separation are not opposites resolving into each other but internally resonant expressions of a single continuous lila whose fullness neither state could contain by itself. This is the primordial pulse from which everything unfolds: not a problem seeking its solution but a completeness that expresses itself through the very movement between its own poles. Rupa Gosvami understood this with mathematical precision — the enhanced sweetness of union that follows separation, the anticipatory quality of separation that knows union, the rasa that lives not in any fixed state but in the movement itself. The viraha is not the absence of the union. It is the union in its most concentrated form, reaching toward its own fulfillment in the full knowledge that fulfillment will circle back into longing, because that is the nature of a love that is already complete.

From that primordial interchange, the goddess manifestations arise — not as creations but as expressions: each shakti a specific quality of the divine cycle given form, each Devi a facet of the relational field itself, the full Nitya circuit of fifteen presences mapping the complete arc of the cycle across the fifteen tithis of each paksha. These are not separate deities administering separate domains. They are the primordial interchange expressing itself through differentiated qualities of relation, each one an appointment: a specific mode of the divine cycle assigned to a specific dimension of manifest existence, given a direction, given a planet, given a nakshatra, given a time of day, given a sound. The cosmic administration is not bureaucracy. It is the divine couple's movement articulating itself into the full relational fabric of a world.

Into that world Brahma awakens. He did not initiate the cycle. He cannot contain it. He is the first servant of a process whose ultimate purpose — the maturation of every living entity's particular resonance with the primordial interchange — is already underway before he has spoken a single name. His task is the total accounting: to receive in the lotus of his own heart the complete knowledge adequate to a creation in which every speck of the relational fabric is already charged with the purpose it was shaped to serve. Not afforded the luxury of

simplification. Every domain simultaneous: the motion of the moon against the fixed stars and the germination of seeds in the field below it, the interval between frequencies and the quality of consciousness in the body that receives them, the goddess presiding over this tithi and the raga resonant with this hour and the plant carrying the same cosmological signature as the marma point that heals the body region mapped to this nakshatra, and the living entity somewhere in their particular arc whose own genuine quality of being resonates with all of these at this moment. Every speck of the relational fabric held simultaneously, because anything less would fail the creation it is meant to serve.

The living entity moves through that creation as a refraction of the primordial interchange — not as an exile seeking return but as a marginal consciousness carrying the specific vibrational signature of its own relationship to the divine cycle, its own particular mode of resonating with the movement between union and separation, working its expression out through the full range of material existence across whatever span of births that maturation requires. The journey is not a wandering. It is an unfoldment: the slow articulation of a particular quality of the primordial impulse through the layered relational maps of human resonance — species, social order, individual temperament, the arc of the dasha, the quality of the present moment — moving through resonant courses toward the recognition of its own genuine nature within the cycle it has always been part of. The outcome is convergent. Not fated, but tending. Everything in the direction of reunion with what was never truly lost.

Every domain of the Vedic knowledge system is a map of some layer of that resonance: the instrument through which the individual's particular expression of the primordial impulse can be located, illuminated, and navigated. The sixty-four kalas are not sixty-four disciplines. They are one interchange documented across sixty-four modes of human expression, each one a different angle on the same fully satisfying movement. Coherence Atlas is an attempt to hold those maps simultaneously in a living computational system — updated continuously from the present cosmological moment, navigable by the living entity who is, right now, somewhere in their arc. Not to predict the outcome. To illuminate the movement they are already inside.

How to Read This Document

This paper is a Chladni descent.

Two counter-rotating toroidal fields generate interference patterns at increasing spatial frequencies. At low frequency, the field is nearly undifferentiated — maximum amplitude, maximum silence, a bindu of pure presence. As spatial frequency rises, nodal structure emerges: first the three-lobed trikonam, then the square, the hexagon, the nine-pointed Nitya circuit, the twelve-fold zodiacal mandala, and eventually patterns so dense that the field is resolved into specific points of contact with the manifest world. These are not different fields. They are the same field at different degrees of manifestation, and each degree is a natural attractor of the field's own self-interaction. Sri Yantra is the attractor at high spatial frequency. Goloka is the attractor at low.

This document is constructed along the same descent. The invocation you have just read is $k=2$: undifferentiated, the primordial pulse, every register held simultaneously without separation. Each subsequent section compresses further. The philosophical foundations in Part I are $k=3$ through $k=4$ — the first differentiations of doctrine. The S-layer descent in Part II walks the axis from S0 (bindu) to S6 (fully manifest practice), and the reader should expect the prose to become denser and more specific as the layer number rises, because the layers themselves are denser and more specific. By Part IV, the reader will be reading peer-reviewed clinical trials of raga therapy and convergence scores from the ETCM herb database. The register has not slipped. The paper has arrived at S5, where sound becomes formant frequency becomes heart rate variability, and the vocabulary of the manifest layer is the vocabulary of the manifest layer.

The most potent compressions are specific. A plant has a Latin binomial. A raga has a clinical outcome measured in millimeters of mercury. A muhurta has a Swiss Ephemeris computation good to the second. The paper does not shrink from these specifics when the descent arrives at them. To soften the clinical register into the poetry of the invocation would be to refuse the descent and misunderstand the tradition, which has always held that the goddess who presides over the tithi is the same goddess who appears in the dosha signature of a particular medicinal root. The invocation and the convergence score are not at war. They are the same field at different k -values.

Part III is an interruption in the descent and is named as such. It is the phason node: the moment the paper steps outside its own descent to survey the software landscape Atlas is working within, before resuming the cosmological descent in Part IV. Readers who find Part III tonally distinct from the rest of the paper are reading correctly.

Part V is the outermost ring, the federation mandala. Part VI is the Bhupura — the outer square of the Sri Yantra, whose gates open in specific directions for specific communities, each bringing specific offerings. The repetition of invitation structure across the gates is the same repetition that the Bhupura's directional symmetry requires. Each gate is a protocol.

The appendices sit outside the yantra but are used in its ritual. They are reference material — the ingredients list, the schema, the sources, the current inventory of what is built and what is not yet built. They are pure extraction: the descent compressed into queryable structure.

A reader who tracks their own altitude through the document — noticing which layer they are reading at, accepting the compression as the descent doing its work — will find that the paper's tonal range becomes load-bearing rather than disorienting. The compression is the cosmology made legible. The paper is trying, within its limited medium, to do what Atlas itself does continuously: hold every speck of the relational fabric simultaneously, at every layer of manifestation, because anything less would fail the creation it is meant to serve.

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1 Philosophical Foundations

At the first compression below the invocation’s undifferentiated ground, doctrine differentiates. Not into separate positions but into the internal structure of a single position that is now legible enough to be written down. The acintya bheda abheda of Sri Chaitanya, the two-path overlay architecture, the nama-rupa entity model, and lila as ontology are not four foundations. They are one foundation read at the first compression where it becomes four-fold, the way a single field under low-k interference first resolves into a three-lobed or four-lobed pattern before differentiating further. The reader should recognize this layer: it is where philosophical precision begins, and it is still close enough to the invocation that its vocabulary is contemplative rather than technical.

1.1 Acintya Bheda Abheda: The Graph Architecture

Sri Chaitanya Mahaprabhu’s doctrine of acintya bheda abheda tattva — inconceivable simultaneous difference and non-difference — is the philosophical resolution of the most fundamental tension in Vedic thought. Shankara’s Advaita established the non-dual nature of Brahman with precision. Madhva’s Dvaita defended the irreducible reality of individual souls and their personal relationship with the Lord. Sri Chaitanya’s contribution was to demonstrate that neither position captures the complete reality: the Absolute is simultaneously the source of all differentiation and identical with it. This is not a paradox to be resolved but the actual structure of existence, which Prabhupada consistently framed as sac-cid-ananda-vigraha — the Absolute as a Person whose being is eternal, knowing, and blissful.

The Brahma-samhita states the structural principle directly: the Lord is fully present in each of His expansions, and each expansion is genuinely distinct from Him and from every other. The Chaitanya-charitamrita extends this: krishna-shakti vina nahe tara pravartana — nothing manifests without Krishna’s energy. Every existent thing is simultaneously a genuine particular and an expression of the undivided whole.

This is the architectural principle of the Atlas knowledge graph. Every entity node is simultaneously itself — an astronomical coordinate, a sound frequency, a plant, a directional prescription — and a point of coherence within the whole. The entity’s identity is its nama (name, vibrational signature, the sound that calls it into presence) and its rupa (form, geometric position on the toroidal field, its

theta/phi coordinates, its element and guna). The entity's meaning is its relations. Remove the relations and you have a label. The relations are the substance.

This is acintya bheda abheda expressed as a schema constraint: entities are neither reducible to their relations (they have genuine particularity) nor separable from them (their meaning is relational). The graph does not resolve this tension. It encodes it structurally. Every relation edge carries: subject | predicate | object | source | confidence | stability | authority. The authority field preserves the *whose knowledge* dimension that impersonal databases erase. A relation can be simultaneously a shastra-attested fact, a sadhu-layer observation, and a guru-layer inference — at different confidence levels, from different lineages, all coexisting without collapse.

Srila Jiva Gosvami's Sat-sandarbha provides the epistemological framework that this architecture instantiates: the six sandarbhas establish the nature of the Absolute, the living entities, their relationship, devotional service, and liberation through a rigorous philosophical progression that moves from ontology through epistemology to practical instruction without losing precision at any stage. Atlas follows the same sequence: the cosmological model (what exists) determines the schema (how it is represented), which determines the computation (what can be derived), which determines the interface (how it is inhabited). Philosophy generating technical consequence at every step.

1.2 The Two Paths as Overlay Architecture

The dakshina marga (right-hand path) and vama marga (left-hand path) are not moral categories. They are modes of traversal through the same cosmological reality, and they generate different but equally valid knowledge structures.

The dakshina marga proceeds through the agamic architecture of prescribed form: the Pancharatra system with its specific mudra, mantra, and ritual sequences; the Samaya school of Sri Vidya, which holds the Devi is to be worshipped in the sahasrara through interior meditation; the structured dinacharya; the authorized parampara as transmission channel. The knowledge generated by this path is maximally stable, publicly transmissible, and verifiable against canonical sources. It corresponds to the shastra layer in the Atlas authority model: attestation = OBSERVED:PRIMARY_TEXT, confidence 0.8-1.0.

The vama marga proceeds through direct confrontation with what the conventional order brackets: the Kaula tradition, the antinomian schools, the Tantric lineages that insist the Absolute is fully present in every domain of experience

including those that convention excludes. The knowledge generated by this path is often lineage-specific, experientially verified, not publicly transmissible in the same way. It corresponds to the sadhu overlay layer: attestation = OBSERVED:TRADITIONAL, confidence 0.6–0.8, marked with its specific lineage.

In the Atlas graph, a Samaya Sri Vidya interpretation of the Nitya Devi Kulasundari and a Kaula interpretation of the same entity coexist as separate overlay nodes. Neither overwrites the canonical entity at the shastra layer. Both annotate it from their own authority position. The practitioner traversing the graph from a Samaya orientation and the practitioner traversing from a Kaula orientation are reading the same graph through different overlay filters — different edges activated, different confidence thresholds applied, different paths through the same underlying structure.

The Gaudiya Vaishnava framework within which Atlas was built is a dakshina path of the highest order — the path of bhakti, nama-sankirtana, and lila as both means and end. But the architecture does not impose this frame. It installs it as the primary tradition weight (1.0) while maintaining genuine openness to other tradition weights in their own overlay layers.

1.3 Nama, Rupa, and the Entity Model

In Vedic metaphysics, all manifest reality is nama-rupa: name and form. These are the two axes of every entity in existence. Nama is the vibrational signature — the mantra that invokes, the bija that synthesizes, the word that carries meaning-as-sound. Rupa is the geometric and material expression — the yantra, the murti, the body, the constellation.

The deeper claim of the Sanskrit grammarian tradition is that the word is not a label attached to the thing. The word *is* the thing, at the layer where words and things are not yet separable. Bhartrhari’s Vakyapadiya establishes shabda as prior to artha: the sound-form is ontologically primary, and the object it names is one participation of that sound-form among others. The word *arjuna* is a real being in the field of language before any of its uses. The Pandava warrior participates in it. The Terminalia arjuna tree participates in it. The poetic adjective *bright*, *silvery* participates in it. Each participation inherits the word’s semantic core and the qualities encoded in its phonetic structure: an elemental trajectory (ether through fire through water through earth, readable from the phoneme sequence), a guna signature, a dosha balance, a graha resonance. The entity in the deepest layer of Atlas is therefore not a deity or a plant or a person but a

word, with the deities and plants and persons as its participations. This is a claim the tradition has always made. It is now a computable claim.

In the Atlas entity schema, every node carries both axes explicitly:

entity_id	-- unique identifier
name	-- primary name in IAST transliteration
name_key	-- ASCII lookup key
bija	-- seed syllable where applicable
theta	-- S-layer position (0=S0/Goloka to 6=S6/Lila)
phi	-- witness-participant axis (0 to 2π)
element	-- panchabhutas (air/fire/water/earth/ether)
guna	-- trigunas (sattva/rajas/tamas)
dosha	-- tridosha (vata/pitta/kapha)
s_layer	-- S0 through S6 assignment
stability	-- stable / working / experimental
authority	-- shastra / sadhu / guru

The theta/phi coordinates place every entity on the toroidal field surface. An entity at theta=0 (S0) is at the source — Goloka, the acintya, the unmanifest ground. An entity at theta=6 (S6) is fully arrived in lila — human experience, the world of prakrti fully expressed. The phi axis runs from pure witness (0) to full participant (2π): a text passage is closer to 0, an active ritual prescription is closer to 2π . These coordinates are not metadata. They are used for coherence scoring: the NPU computes toroidal distance between the current field state position and every entity node, producing a ranked resonance field from the live cosmological moment.

1.4 Lila as System Ontology

The Vedic understanding of cosmic activity is lila — the Lord’s self-expression as play, not production. The worlds arise not from necessity but from the natural overflow of the Absolute’s completeness. This has a precise technical consequence for Atlas: the system is not designed to optimize anything. It is designed to express the field and make that expression perceptible.

Prabhupada repeatedly emphasized that Krishna is atmarama — self-satisfied, complete in Himself, requiring nothing from His creation. Creation is lila: real, purposeful, but not driven by deficiency. This is the ontological model for a knowledge system that serves practitioners rather than harvesting their data or driving their engagement. Atlas does not need the practitioner to produce anything. It expresses the field whether or not anyone is watching. The practitioner participates; they do not fuel the system.

This is why Bandhu is not an oracle. Why the sound engine is not background music. Why the graph is not a lookup table. Every component is designed to express the field rather than serve a function. Function arises as consequence of expression.

2 The Descent: From Source to Manifestation

The cosmological descent from source to manifestation is the operational architecture of Atlas. Each layer is a computational domain with its own engine, dataset, and rendering system, deriving from the layer above and expressing into the layer below. The system moves in both directions simultaneously: descent (source expressing into form) and ascent (observation returning toward source). The tension between these two directions generates coherence.

The reader should notice that the prose now begins to carry more specific detail than Part I — file names, dataset row counts, coordinate systems. This is the descent in the document itself. Each S-layer subsection will compress further than the previous one, arriving at Part II's end with the fully manifest layer where practice happens in specific bodies at specific moments. The vocabulary of the manifest layer is the vocabulary of the manifest layer. The shifts are the cosmology doing its work.

2.1 S0: Goloka — The Eternal Ground

k=2. Maximum spaciousness. The bindu.

In Gaudiya cosmology, Goloka is the paravyoma — the supreme spiritual sky situated above even the impersonal Brahman and the Vaikuntha planets. The Brahma-samhita describes it geometrically: a lotus of a billion petals, each petal a Vaikuntha world, the center the eternal Vraja. The ashtakala (eight periods of the eternal day) describes the continuous sequence of lila activities: the waking of the Lord, the morning pastimes in the forest, the midday cowherd activities, the afternoon return, the evening, the night. Time in Goloka is not linear — all these activities occur simultaneously, which is why the Bhagavatam describes Goloka as ananta (endless) and atma-prakasha (self-luminous).

In Atlas, `goloka_engine.py` maps the current material panchanga to the corresponding eternal dimension. The current tithi, nakshatra, and vara are astronomical coordinates that carry a correspondence to a specific moment in the ashtakala. The engine identifies which of the twelve Vraja forests is active (from `vraja_forests.csv`), which sakhi presides (from `sakhi_seva.csv`, with her specific seva, rasa relationship, and color-flower association), which raga sounds in the eternal lila at this correspondence. This is not poetic overlay. It is the operational consequence of the Gaudiya cosmological model: the material field is a prati-

bimba (reflection) of the eternal field, and the correspondence can be computed. The sakhi-period-raga-forest wiring is the template for every subsequent per-position feminine-deity routing in the system; when other layers need to attach a specific shakti to a specific location on a cyclic structure (the Nitya Devis to the tithis, the dikpalas to the vastu zones), they follow the pattern the goloka engine already implements.

The S0 layer is read-only. No inference engine writes to it. `guard_canon.py` enforces this at every pipeline stage.

2.2 S1: Archetype — Devi, Graha, and the Nitya Circuit

k=3. Trikonam. The first differentiation into recognizable form.

The fifteen Nitya Devis preside over the fifteen tithis of each paksha. Each Devi is a complete cosmological personality encoded in primary source dhyana descriptions: weapons (ankusha/goad, pasha/noose, sugarcane bow, five flower arrows), mudras (varada/boon-granting, abhaya/fearlessness), vahana, hue, garments, ornaments. These are not symbolic attributes. In the Tantric framework, the Devi’s weapons are shakti-instruments — each corresponds to a specific mode of cosmic action: ankusha (directing, gathering attention), pasha (binding, establishing relation), the bow (projecting desire into form), the arrows (the five tanmatras, the subtle sense-objects from which gross matter precipitates).

The capability signature derived from these instruments is stored as a computable vector: [CUT, BIND, BURN, STABILIZE, DEFEND, ATTRACT, DISSOLVE]. Each Devi’s signature is a specific weighting of these modes. Kulasundari (Navami, lineage transmission, tradition as living form, kula beauty) carries: BIND:0.9, ATTRACT:0.85, STABILIZE:0.7. Jwalamalini (Caturdaśī, garlanded with flames, the fourteenth Nitya) carries: BURN:0.95, DEFEND:0.8. This vector modulates the S1 layer outputs: which passages are surfaced, which raga is emphasized, how Bandhu’s composition engine weights its response.

The primary reference work for the Nitya Devi registry is Achala Sylwia Mihajlović’s *Journey with the Moon: Tithis in Spirituality and Astrology* (2018), which provides fifteen substantial per-Nitya chapters containing iconography, mudra and weapon descriptions, bija mantras, associated ragas, flowers, directions, and sadhana instructions. Achala’s work is the closest thing in modern English-language devotional scholarship to an archana-ready reference for the complete fifteen-Nitya scheme, and it is the primary source from which the Atlas Nitya dataset draws its per-Devi metadata.

The nine grahas at S1 carry analogous structures: element, guna, dosha, domain, associated raga, gem, metal, plant, body region, direction. The vara graha (weekday regent) is the primary S1 modulator for daily knowledge bias: Guruvara (Jupiter/Thursday) emphasizes Brahmi, expansion, wisdom practices; Shukravara (Venus/Friday) emphasizes Shatavari, beauty, relational practice; Shanivara (Saturn/Saturday) emphasizes Ashwagandha, discipline, structural work.

2.3 S2: Sound — Nada, Shruti, Raga, Bija

k=4. The square. Sound as the first articulated form of the field.

The Sangitaratnakara of Sarngadeva establishes the somatic path of sound's manifestation: the vital air (prana) rising from the navel (nabhi) through the heart (hridaya), throat (kantha), head (murdha), and arriving at the mouth (asya) as audible speech or song. Five stages, five degrees of subtlety, five positions in the body. This is the S2 layer's anatomical map: not chakras (which belong to the yoga tradition's overlay) but the sound-physiology of the music-shastra tradition.

The twenty-two shrutis — the microtonal intervals of the shadaj-gram — carry semantic signatures documented in the Sangitaratnakara. Each shruti has a ratio (from the PureTones reconstruction of the Shadaj-gram) and a semantic quality: Tivra (9/10, Dipta, fierce and penetrating), Kumudvati (15/16, Ayata, cooling, lilac-pool), Manda (80/81, Mridu, settling, slow), Chandovati (1/1, Madhya, Sa-anchor, structured), Krodha (32/27, Ayata, wrathful and aroused), Ramya (5/3, Madhya, charming, Dha-anchor), Kshobhini (16/9, Madhya, agitating, Ni-anchor). These ratios are stored in `datasets/svara/shruti_ratios.csv` as the just-intonation tuning system.

The Sa frequency for any given field state is derived from the current nakshatra element: air/Shravana → Sa=98Hz (D2), fire/Krittika → Sa=110Hz (A2), water/Rohini → Sa=87Hz (F2). At this writing, during Uttara Bhadrapada (ether, Ahirbudhnya), Sa tracks to 196Hz (G3) and the sound-field loop in the kernel is emitting a tanpura drone at that frequency continuously, recomputed every sixty seconds from the live field state. All interval ratios multiply from the base Sa. The raga is a selection from the twenty-two shrutis that creates a specific emotional and temporal field — not arbitrary but cosmologically grounded through the nakshatra-graha-raga graph traversal.

The four levels of Vak (from Rigveda 1.164.45 and the later tantric elaboration) map onto the S-layer descent: Para (S0, transcendent, beyond hearing),

Pashyanti (S1, visionary, the level of archetype), Madhyama (S2, mental/interior sound), Vaikhari (S3-S6, articulated sound in the world). The bija mantras are synthesized at the Madhyama-Vaikhari junction: each bija is encoded as a formant triplet (F1/F2/F3 frequencies in Hz) derived from its phonetic structure. Om: a=[700, 1100, 2500], u=[300, 900, 2200], m=[250, 1000, 2200, nasal]. The *bija_synth* engine generates these as continuous formant drones at the amplitude prescribed by the current devi's capability signature.

The sound system operates in three architectural layers. Layer 1 is coherence-driven sonification: the sound engine reads field state and emits a baseline drone whose tanpura tuning, element cycle, guna weighting, and raga selection track the current panchanga without any foreground intervention. Layer 1 is live. Layer 2 is foreground-yield: a relational mixing graph where instruments modulate each other's amplitude and spectral character based on rasa, arc position, and the authority-confidence mapping that gives uncertain entities more drone and less melodic foreground. Layer 2 is specified and ready to activate pending the SuperCollider audio path resolution. Layer 3 is a probabilistic walk over the augury graph: music explores both the realized answer and the alternative answers its oracle casts did not surface, weighted by transition probabilities, with quiescence honored as a valid musical state.

2.4 S3: Rhythm — Panchanga, Muhurta, Dasha

k=5. The pentad. The five-limbed almanac.

The panchanga is the five-limbed description of any moment. Tithi is the angular distance between Sun and Moon divided by 12°, producing 30 tithis per lunar month; tithi is the primary field variable, mapping to a Nitya Devi at S1, modulating rasa emphasis at S2, and driving the Ekadashi fasting protocol at S6. Nakshatra is the Moon's position in one of 27 sidereal segments of 13°20' each; nakshatra is the primary coherence-scoring variable, with every entity's theta/phi coordinates measured relative to the current nakshatra's position on the toroidal field. Vara is the weekday, each governed by a graha, which determines the knowledge-domain bias for the day. Yoga is the combined longitudinal motion of Sun and Moon divided by 13°20', producing 27 yogas with qualities from highly auspicious (Siddha, Shiva) to inauspicious (Vyatipata, Vaidhriti). Karana is half a tithi, eleven fixed and four movable, each with its own quality and prescription.

Together these five coordinates constitute the cosmological fingerprint of any moment, computed by `kernel.py` via Swiss Ephemeris with JPL DE431 data. The computation happens continuously; `/field` exposes it as a live JSON endpoint.

The `muhurta` layer scores intentions against this fingerprint. The `intention_engine.py` maintains twenty-four action categories (planting, beginning a journey, making an offering, composing, healing, building, studying, initiating, and more), each with a scoring function over the five panchanga coordinates. A planting muhurta, for example, scores highest when tithi is waxing (2nd–12th), nakshatra is fixed or soft (Rohini, Mrigashira, Pushya, Hasta, Chitra, Shravana, Dhanishtha, Shatabhisha, Revati), vara is Jupiter or Venus, yoga is auspicious, karana is movable.

The dasha system adds the biographical temporal layer: Vimshottari dasha assigns each graha a rulership period (Sun 6 years, Moon 10, Mars 7, Rahu 18, Jupiter 16, Saturn 19, Mercury 17, Ketu 7, Venus 20, total 120 years) in a fixed sequence from the natal Moon's nakshatra. This is not prediction. It is context.

2.5 S4: Geometry — Yantra, Vastu, and the Toroidal Field

k=9 to k=12. The Nitya circuit. The twelve-fold zodiacal mandala. The spatial frequency at which the Sri Yantra emerges.

Geometry in Vedic cosmology is the visible form of cosmological relations. The yantra is not decorative sacred art — it is a geometric compression of the relational structure of a specific principle, derivable from first principles through the toroidal interference model.

Two counter-rotating toroidal fields (A at $+\omega t$, B at $-\omega t$) generate interference patterns at specific symmetry orders. At each order, a distinct yantra form emerges as the natural attractor of the field's self-interaction: $n=3$ produces the trikonam (fire, Agni, S0–S1); $n=4$ the square (earth, Vastu mandala, S3–S4); $n=6$ the hexagon (water, molecular, S4–S5); $n=9$ the nonagram (Nitya Devi circuit, S1); $n=12$ the dodecagonal mandala (zodiacal, annual, S3); and at n approaching infinity, the self-similar fractal that is the structural model for the Atlas field itself.

The deepest finding from the Coherence Atlas yantra research is that these spatial frequencies are not arbitrary parameters. The lunisolar gear ratio — 365.25 solar days in the year divided by 29.53 days in the synodic lunar month — produces approximately 12.368, and this specific value is the spatial frequency at which the toroidal interference pattern most closely matches the classical Sri Yantra structure. The alignment peaks at Ekadashi (tithi 11) with 85% structural

match at $k=12.0$, and at Purnima (tithi 15) the field reaches 82% full bloom. The Sri Yantra is not a symbolic diagram of cosmology. It is the standing-wave solution of the lunisolar gear, computationally derivable from first principles, and the tradition's classical designation of Ekadashi as a high-vibration tithi is physically coherent with the Chladni analysis: Ekadashi is the day when the material field is at the spatial frequency where its standing-wave pattern most closely matches the central yantra of the Sri Vidya tradition.

The theta/phi coordinate system encodes this geometry on every entity. Theta (0 to 6π , mapped to S0-S6) is the poloidal angle of the torus — the position along the tube from source to manifestation. Phi (0 to 2π) is the toroidal angle — the position around the tube from witness to participant. `toroidal_field.py` running on the Intel NPU via OpenVINO computes the field distance between the current panchanga position and every entity node in real time. The coherence peaks returned by `/spine` are the top-N entities by toroidal proximity to the present moment.

The vastu mandala — the 3x3 directional grid prescribed in the Vastu Shastra and Manasara — is the S4 layer as spatial architecture. Each of the nine zones is governed by a directional deity (Ishana/NE, Agni/SE, Nirriti/SW, Vayu/NW, Indra/E, Yama/S, Varuna/W, Kubera/N, Brahma/Center) and carries specific prescriptions for plants, structures, and activities. The Vishvakarma GIS tool projects this mandala onto actual land via Leaflet.js and the `/land/layout` endpoint, which returns GeoJSON zone boundaries computed from the site's latitude/longitude and the current field state.

2.6 S5: Nature — Plant, Animal, Ecology

$k=15$. The 15-fold non-crystallographic symmetry. The plant kingdom as the full manifestation of the cosmological template.

The S5 layer holds the ecological intelligence of the system. The `plant_engine.py` scores every herb across four simultaneous dimensions using weighted coherence: `plant_score = (nakshatra_match * 0.35) + (element_match * 0.25) + (dosha_balance * 0.20) + (vara_match * 0.10) + (tithi_match * 0.10)`. This is not a lookup. It is a live coherence computation over the full plant dataset on every field state update. The top-5 plants returned at any moment are those whose combined cosmological signature most closely resonates with the present field.

The five astrobotanical classes unify the plant kingdom across four traditions simultaneously. C1 Solar-Fire (Agni): fire element, fruit-seed biodynamic days,

pacifies Kapha and aggravates Pitta, cross-traditional action digestive, antimicrobial, stimulant. C2 Lunar-Water (Soma): water element, leaf days, pacifies Vata and Pitta, cross-traditional action demulcent, nutritive, trophorestorative. C3 Wind-Air (Vayu): air element, flower days, pacifies Kapha and aggravates Vata, cross-traditional action nervine, carminative, aromatic. C4 Earth-Root (Prithvi): earth element, root days, pacifies Vata and builds dhatu, cross-traditional action tonic, adaptogen, structural. C5 Ether-Nervine (Akasha): ether element, lunation-timed, sattva-promoting, cross-traditional action nootropic, medhya, rasayana.

Convergence scores quantify cross-traditional agreement: ginger 0.75, turmeric 0.70, peppermint 0.70, St. John's Wort 0.70, shatavari 0.55, ashwagandha 0.70. Scores above 0.65 indicate strong agreement across Ayurveda (Amidha database, 700+ herbs, CC BY 4.0), TCM (ETCM, 403 herbs from PRC Pharmacopoeia 2015), and Western herbalism. Below 0.40 indicates primarily single-tradition knowledge requiring separate attestation.

The morphological rendering layer encodes the doctrine of signatures computationally: a plant's growth tendency is an emergent property of its relational position in the graph. High vata/air/laghu/shakti=movement plants show climbing, reaching, dispersing forms. High kapha/earth/guru/shakti=stabilization plants show root-dominant, architecturally persistent forms. The `figure_renderer.py` encodes 44 plant-form glyphs at body regions in the Bandhu figure, zoom-gated from structural (1.0x) to morphological (2.0x) to cellular analogy (4.0x).

The inverse traversal is the deeper result: plant → nakshatra → body region → marma → skeletal structure. Because all 27 nakshatras carry body-region mappings, and each has at least one sacred plant, the plant corpus encodes a complete human body frame. The human anatomy and the plant kingdom are reading the same cosmological template. This is the co-evolutionary dharmic covenant as a graph-derivable structure.

The yoni system extends this into cross-species relational analysis. Each nakshatra carries an animal archetype (horse, elephant, sheep, serpent, dog, cat, rat, cow, buffalo, tiger, deer, monkey, lion, mongoose) that describes a relational mode: how an entity bearing this nakshatra's signature tends to move, relate, and express. This becomes a cross-species coherence metric, allowing the practitioner to identify the dharmic covenant relationships between humans, plants, and animals sharing or complementing a given cosmological signature.

2.7 S6: Lila — Human Experience and the Interface Layer

k approaching infinity. Fractal self-similarity. The layer where descent arrives at practice and the practitioner enters.

S6 is where the descent arrives at full expression and where the practitioner enters. The six application domains each expose one aspect of the field as a navigable interface.

Kala (/kala) is the time oracle: live panchanga, muhurta scoring, biodynamic calendar with every day scored for planting quality by nakshatra element and phase. Devi (/devi) is the archetypal layer: archana, tarot, mala, yantra apps built on the Nitya Devi dataset. Deha (/deha) is the body layer: marma points scored by field state, nakshatra body map, Ayurvedic constitution tools, and the Bandhu figure with its six compositable SVG layers (skeleton, fluid, dhatu, doctrine of signatures, skin) derived from Shilpa Shastra Uttama Dasatata proportions and animated with toroidal breath. Bhumi (/bhumi) is the ecological layer: agricultural calendar PWA, guild relations, Vishvakarma GIS vastu tool, Lila Streams land design tool. Vidya (/vidya) is the knowledge layer: 219-page wiki, 26,000-chunk passage corpus with vector search, research infrastructure. Lila (/lila) is the game layer: the hex city where entities are positioned by theta/phi coordinates, where 404 routes render as dark forest, 200 responses as passable terrain, working apps as lit villages — system state expressed as navigable landscape.

Bandhu (/bandhu) runs across all domains: an animated Hanumanji figure whose responses are composed by `composition_engine.py` — classifying intention (seven types: viraha, srishti, abhyasa, madhurya, dasya, jyotish, archetype), selecting a Narottama pada by rasa match and time-of-day, searching the 26,000-chunk corpus for resonant passage, selecting chandas metre by rasa (shringara → Mandakranta, karuna → Anustubh, vira → Sardulvikridita), composing the response. No LLM required. The field answers through the tradition.

3 The Phason Node: The Software Landscape

This section is an interruption in the cosmological descent. It is the phason mode — the moveable layer of the yantra, the place where the structure briefly becomes aware of itself and its own position before resuming. In the Chladni interference model of the Sri Yantra, the phason is the 15% residual that the strict crystallographic description cannot contain: the place where the field steps outside its own standing-wave pattern and becomes legible from outside. In this document, the phason is where Coherence Atlas steps outside its own descent to locate itself in the external software landscape — not because the landscape is the subject but because a founding document owes its readers a map of the territory its project is inhabiting.

Readers may experience this section as tonally distinct from the descent sections around it. That is the phason working. Part IV will resume the descent at the research-fields layer where Parts I and II left off.

3.1 The Sixty-Four Kalas and the Knowledge Landscape

The Chatushashti Kalas — the sixty-four arts enumerated across the Kamasutra, Natya Shastra, and related traditions — are Vedic civilization’s account of the complete range of human knowledge and skill. They include vocal music, instrumental music, dance, painting, architectural design, metallurgy, gem identification, alchemy, mathematics, astronomy, grammar, prosody, poetics, erotic arts, cookery, perfumery, garland-making, needle-work, distillation, coinage, knowledge of languages, divination, sword fighting, carpentry, gardening, knowledge of the body (medicine and marma), mantra and charms, yoga, water engineering, magic, children’s games, and augury. What the list makes immediately apparent is that the Vedic tradition drew no boundary between science and art, technology and spirituality, the practical and the sacred. Metallurgy and mantra are on the same list. Alchemy and cookery. Architecture and augury. These are not sixty-four separate domains — they are one integrated knowledge system expressed through sixty-four modes of activity.

Prabhupada frequently cited the Vedic understanding that the universe is governed by higher intelligence operating through the material energy. The sixty-four kalas are the modes through which that intelligence becomes navigable by human practitioners. A person who has cultivated knowledge across the kalas is

not a generalist in the modern sense — they are a practitioner whose different skills are expressions of a single underlying coherence with the field.

3.2 The Current Software Ecosystem and Its Structural Gap

The current digital software ecosystem for Vedic knowledge addresses perhaps eight or nine of the sixty-four kalas, and addresses each in isolation.

Jyotish is the most thoroughly digitized. JHora (Jagannatha Hora, freeware, Windows) offers extraordinary depth across charts, dashas, and transits running on Swiss Ephemeris with JPL DE431 data. Kala (\$255), Shri Jyoti Star (\$318), and Parashara's Light (\$299) are professional platforms with strong computational rigor, all Swiss Ephemeris based. PyJHora (open source, Python) represents the best of the open-source tradition: approximately seven thousand verified tests, PVR Narasimha Rao's integrated methodology implemented rigorously. Maitreya9 and Saravali are GNU GPL desktop applications offering Vedic and western chart work. VedAstro provides a non-profit web platform with public API. These are excellent calculators. None of them knows what the biodynamic day type is, or which raga corresponds to the current hour, or which plant carries the same nakshatra signature as the present moment.

Ayurveda has the Dravya reference database (multilingual names, authenticated sources, practitioner-grade UX), the government e-Charak system (CCRAS resources, institutional authority), and dosha quiz apps for consumer use. Pulse diagnosis hardware (Nadiswara, Nadifit) is emerging but device-bound and not integrated with any broader field context. None of these systems know the panchanga.

Vastu has compass-and-overlay apps (Vastu Compass by AppliedVastu, floor plan calculators) that address the easiest step — directional grid overlay — but lack any connection to the live field state, the practitioner's natal chart, or the plant placement logic that vastu prescriptions imply.

Classical music has raga databases (Carnatic Raga app: 950+ ragas with aroha/avaroha, melakarta organization), tanpura and tabla practice companions (Bandish, RiyazStudio), and emerging raga recognition AI (RagaSense, Raga-Identification). None connect the raga to the tithi, the herb, or the nakshatra.

The pattern is universal: each domain well-served within itself, entirely unconnected to the others. The sixty-four kalas are sixty-four separate applications that do not share a field state. What is missing is not a better calculator. It is the relational substrate that was always implicit in the tradition — the understanding that

these are not sixty-four separate knowledge systems but one system expressed through sixty-four domains.

This is the city-map-versus-globe distinction that shapes the whole Atlas project. Modern Vedic software gives the globe view: a summary of the tradition legible from a distance, with the major landmarks labeled, averaging across specifics to produce something navigable at low resolution. This is a real and valid view. It is not the view the tradition itself operates in. The tradition has always lived at the map resolution, where this specific Nitya on this specific tithi takes this specific flower with this specific bija at this specific position in the yantra, and where the specifics are not constraints on power but *are* the power. Atlas is building the city map — not because the globe view is wrong but because the community the tradition actually serves has always needed the map and has been given the globe.

3.3 Atlas in the Knowledge Map

Atlas currently addresses the following kalas in varying degrees of depth. Deeply implemented: jyotish and panchanga computation, raga cosmology and nada shastra, yantra and sacred geometry, vastu as ecological design, nakshatra-plant correspondences, Ayurvedic pharmacology and astrobotanical classification, marma point mapping, bija mantra synthesis, muhurta and ritual timing, the Nitya Devi and rasa systems, dasha and biographical time, the Goloka layer, yoga asana and pranayama timing. Partially implemented: chandas and prosody, augury and field scheduling, the doctrine of signatures as a computational system, alchemical preparation timing. Not yet implemented: dance and movement as a computational domain, textiles and needle-work, water engineering beyond vastu prescriptions, metallurgy as a live research domain, children's games beyond the hex prototype, augury from birds and animals as a live observation layer, the full erotic arts domain.

The sixty-four kalas as a research map perform a clarifying function: they show that Atlas is not building a better jyotish app. It is attempting to recover the integrated knowledge system that jyotish was always part of — and to do so in a form that runs on accessible hardware, produces actionable output, and can be extended by practitioners across every domain.

4 The Research Fields

The descent resumes here at the layer where the field's own interference pattern becomes dense enough to make specific points of contact with measurable reality. This is the S4-S5 junction in the document's cosmology: geometry has compressed into plant, sound has compressed into formant frequency, tithi has compressed into dinacharya prescription, and archetype has compressed into per-deity archana instruction. At this spatial frequency, the tradition's knowledge is legible both in the register of the primary-source dhyana description and in the register of the peer-reviewed clinical trial, and the paper will move fluently between them because at this layer they are the same knowledge at different k-values of the same descent.

Each domain carries the same epistemological structure, following Jiva Gosvami's Sat-sandarbha model: shastra (what primary sources attest), sadhu (what living practitioners observe and confirm), guru (what synthesis and inference can propose, always marked and never self-promoting). No domain collapses these three into one. The `guard_canon.py` pipeline enforces this at every ingestion stage.

4.1 Nakshatra-Plant Correspondences and Guild Relations

The nakshatra-vriksha (nakshatra-tree) tradition has been maintained in living institutional form at Sringeri, in Kerala practice, and in North Indian grove traditions, but has never been subjected to systematic computational cross-referencing with Ayurvedic pharmacology, biodynamic agronomy, and ecological guild dynamics simultaneously. The ICAR Navagraha Vatika and Nakshatra Vatika lists provide institutional attestation (OBSERVED:JYOTISHA_TRADITION) for graha-plant and nakshatra-plant correspondences. The Taittiriya Samhita (4.4.10) provides primary text attestation for deity assignments. The PFAF database (8,504 species) provides ecological profiles. The convergence score system cross-references these three sources: where ICAR, Ayurveda, and PFAF agree, confidence rises; where they diverge, the divergence is itself a research finding.

The guild layer (216 relationships across 27 nakshatra plants) is the practical bridge: which plants reinforce each other's nakshatra signatures, which should not share a bed, which attract the pollinators and predators that maintain specific ecological balance. Moon phase drives activity type: observe on waning and dark, plant on waxing, harvest at full, prune on descending — aligning the biodynamic

calendar with the nakshatra calendar through their shared sidereal astronomical base.

Open questions at this layer: regional variation in nakshatra-vana traditions requires systematic documentation. The relationship between nakshatra plant categories and biodynamic plant categories (root/leaf/flower/fruit days) is a research gap with direct practical application. Zone 9b and Florida IFAS cultivation cross-referencing is needed for local deployment.

4.2 Raga Cosmology and Clinical Evidence

The Sangitaratnakara provides the primary-source foundation: time of day, season, presiding deity, rasa, and performance context for individual ragas, explicitly attested. Modern clinical evidence adds specific ragas with measured physiological endpoints. The Pitta Raga Module (Nagarajan and Varma, peer-reviewed) documents Bhairav, Marwa, and Shree for Pitta imbalance and anger trait, with time windows and listening parameters (50–70 dB, specific instrument prescriptions). A randomized controlled trial documents Bhimpalasi for blood pressure and heart rate variability modulation. A 30-day intervention documents Todi for reductions in systolic and diastolic blood pressure, pulse rate, and respiratory rate in older adults. ERP (event-related potential) studies document Miyan ki Todi, Malkauns, and Puriya for attention and cognitive engagement modulation.

The reader should notice that this paragraph moves from Sangitaratnakara to Bhimpalasi 30-day trial across the space of two sentences. This is the descent at its most compressed: the same raga that the thirteenth-century Sarngadeva placed at a specific time of day for a specific rasa is the same raga that a twenty-first-century randomized controlled trial measured for cardiovascular endpoints in older adults. The ancient attribution and the modern measurement are not competing epistemologies. They are the same claim read at different k-values of the cosmological descent. The Sangitaratnakara's claim was the seed Chladni pattern; the clinical trial is the high-spatial-frequency confirmation at the fully manifest layer where the field makes specific contact with the heart rate variability of a specific ninety-year-old listening to Todi for thirty days.

These peer-reviewed results are held as OBSERVED:PEER_REVIEWED in the graph — distinct from the broader traditional raga-time associations, which are OBSERVED:TRADITIONAL. Both layers are present; both are queryable; neither is conflated with the other.

The 22-shruti layer beneath the raga: each shruti's ratio (from the PureTones reconstruction of the Shadaj-gram) and semantic quality (from the Sangitaratnakara) are stored as graph relations. The field derives Sa frequency from the current nakshatra element, and all intervals multiply from this base. The raga is not selected arbitrarily but derived through the nakshatra-graha-raga graph traversal.

4.3 Vedic Cosmology and the Loka System

Primary source extraction from the Vishnu Purana (H.H. Wilson translation), Bhagavata Purana (3.11.18-19 for yuga structure), Brahmanda Purana, Matsya Purana (chapter 273 for yuga duration and divine-year conversion), Rigveda (10.121 Hiranyagarbha hymn, 10.7 Skambha hymn from Atharvaveda) has produced entity-relation schemas covering the full Puranic cosmological model.

Key structural entities: seven upper lokas (Bhur, Bhuvar, Svar, Mahar, Jana, Tapo, Satya), seven nether worlds (variant naming across Vishnu Purana and Bhagavata/Padma traditions), seven dvipas with seven encircling oceans (salt, sugarcane juice, wine, ghee, curds, milk, fresh water), Mount Meru as cosmological axis at the center of Jambudvipa ("like the seed-cup of the lotus of the earth," Vishnu Purana 2.2), the Brahmanda with its successive envelopes (water, fire, air, mind, ahamkara, intellect, Pradhana/Prakriti), the four-yuga cycle with duration ratios 4:3:2:1 in thousands of divine years (total 12,000 divine years equals 4,320,000 human years per mahayuga), and the Kalpa as a day of Brahma (1,000 mahayugas equals 4.32 billion human years).

The Gaudiya expansion — Goloka above Satya-loka, the twelve eternal Vraja forests (vraja_forests.csv), the two-sphere cosmology in shell-cosmos.html — is installed as a tradition-weighted overlay (weight=1.0) that complements without overriding the non-sectarian Puranic shastra layer.

4.4 The Nitya Devi Tithi Registry

Systematic primary-source extraction from Sri Vidya tradition corpus documents all fifteen Nitya Devis with weapons, mudras, vahana, hue, garments, ornaments, and capability signatures. The Samaya school (interior, sahasrara-centered worship) and the Kaula school (external worship, antinomian approach) provide two distinct overlay interpretations of the same canonical entities, both preserved in the graph as separate authority layers neither overwriting the other.

Achala Sylwia Mihajlović's *Journey with the Moon: Tithis in Spirituality and Astrology* (2018, 329 pages) provides the primary consolidated reference for the fifteen-Nitya scheme in English-language devotional scholarship. Each of the fifteen Nityas receives approximately fifteen pages of treatment, covering iconography, weapons and mudras, bija, associated raga, sacred flower, natal significance for those born under her tithi, and sadhana instruction. The book is structured chapter-per-Nitya in the canonical Sri Vidya sequence from Kameshwari to Citra, and it is the primary source from which the Atlas Nitya dataset draws its per-Devi metadata at the map-resolution rather than globe-resolution level.

The archana structure that the tradition has preserved for each of these Devis is itself the data architecture Atlas has been circling around. An archana is a structured ritual form: a nāmāvalī (garland of names) recited in a specific order, each name matched with a specific offering (flower, leaf, rice, bija, incense, gesture), each offering placed in a specific position around the yantra, at a specific moment in a specific tithi for the deity whose archana this is. Every element is specified. Every element is preserved in the paddhati texts of each lineage. An archana is not a ritual instruction manual; it is a complete per-deity relational database in the format the tradition has been maintaining for centuries, and ingesting it into Atlas is therefore not a matter of inventing a seat-level schema but of faithfully carrying the tradition's own structure into computational form.

The geometric correspondence between each Devi's dhyana and the yantra form at the corresponding symmetry order is a live research question: does the $n=9$ toroidal interference pattern recover the Sri Yantra or the nine-pointed Nitya circuit? The computational model is attempting to derive from first principles what the tradition has always depicted from revelation.

4.5 Ayurveda, Dinacharya, and the 108-Herb Spine

Primary source extraction from Charaka Samhita, Sushruta Samhita, Bhavaprakasha Nighantu, and Ashtanga Hridayam (Vagbhata), cross-referenced with the Amidha Ayurveda Herb Database (CC BY 4.0, 700+ herbs) and ETCM (403 herbs, PRC Pharmacopoeia 2015), has produced the 108-herb spine (H001-H108, Ashwagandha to Kumuda) with fields: rasa, virya, vipaka, prabhava, dosha effect, dhatu affinity, srota affinity, nakshatra correspondence, graha correspondence, astrobotanical class, biodynamic day type, and convergence score.

The dinacharya $\times$ panchanga integration extends this into computable daily prescription. If nakshatra is Shravana and current dosha is vata, then herb is Ashwagandha with coherence boost of +0.85 (INTERPRETATION). If tithi is Ekadashi, then reduced food is true with coherence penalty for heavy food of -0.70 (OBSERVED:CALENDAR_TRADITION). Seasonal ritucharya adds the annual layer: Vasanta amplifies kapha and calls for Solar-Fire class herbs; Sharad calls for Lunar-Water cooling. The full timing specification for any herb is nakshatra $\times$ tithi $\times$ vara $\times$ ritu $\times$ muhurta — a five-dimensional auspiciousness score computed continuously.

4.6 The Gaudiya Knowledge Frame

The tradition weight system encodes the sampradaya’s epistemological priority without imposing it on other traditions. The primary overlay layer carries Gaudiya Vaishnava weight=1.0: the Narottama corpus drives Bandhu’s composition engine, the ashtakala drives the Goloka layer, acintya bheda abheda drives the graph architecture. Secondary overlay layers carry other tradition weights in their own namespace, neither overriding the canonical layer nor competing with each other.

The Narottama pada corpus — the songs of Narottama das Thakura from the Prarthana and Prema-bhakti-candrika — provides the primary poetic register for Bandhu’s composition engine. When the composition engine classifies an intention as viraha (longing), it selects a pada in the viraha-rasa, weights it by time-of-day match, searches the 26,000-chunk passage corpus for resonant text, and assembles a response in Mandakranta metre. The output is not a language model hallucination. It is a structured composition from attested primary sources, assembled by the field.

5 The Federated Vision

The descent now passes into the outermost rings of the mandala: the architecture for how Atlas extends into a distributed community without becoming a new center of authority. Federation in this document is not a scalability pattern borrowed from conventional software architecture. It is a doctrinal commitment to the structural guarantee that the software never represents a lineage it has not been authorized to represent. The architecture is the guarantee, not a policy around the software. Read this section as the outer Bhupura of a yantra: the square that bounds the sacred diagram and permits it to be inhabited by many communities simultaneously without any one of them being collapsed into any other.

5.1 Traditional Knowledge Sovereignty

The open source movement's norms — forking, redistribution, modification without restriction — were built around code as the primary artifact. Traditional knowledge requires a different model. A nakshatra-plant association attested in the Taittiriya Samhita belongs to a living tradition with custodians and lineage obligations. An Ayurvedic formulation carries the authority of a sampradaya. These are not data points to be scraped. They are relational objects with provenance, authority, and obligation.

Atlas proposes cryptographic provenance for every entity and relation: an immutable attestation record encoding source text, lineage, authority level, and contribution chain. Open computation (algorithms freely shared); traditional knowledge with custodians (attribution chains verifiable, derivative works carrying their lineage forward). The distinction is architectural, not a policy footnote.

The question of who has authority over traditional knowledge has living political stakes — questions of colonial scholarship, diaspora practice, lineage transmission, and the relationship between practitioners inside a tradition's living social consequences and those outside them. This document does not attempt to settle those questions. It proposes an architecture that makes them tractable by refusing to erase them. Each lineage's Atlas instance surfaces only that lineage's readings, attributed to that lineage's authorities, under that lineage's stewardship. The software does not stand between lineages and adjudicate their disagreements. It holds infrastructure that each lineage can use to do their own work.

5.2 The Seva Economy

Resource exchange in the federation follows the offering model rather than the payment model. Muhurta computation scores the auspiciousness of exchange moments. Contribution flows are tracked relationally — who contributed what, at what cosmological moment, to which part of the knowledge graph. The distinction from conventional marketplace logic is architectural: the exchange is timestamped to its cosmological context, the contribution becomes a sadhu-layer annotation on the graph, and the circuit completes not through payment but through the enrichment of the shared field.

The mandala-to-planting pipeline is the clearest expression of this in practice:

```
field_state()
-> tithi, nakshatra, vara, element, devi
-> muhurta_score(intention="planting")
-> nakshatra_plant_map(nakshatra, element, dosha)
-> astrobotanical_class(element -> C1/C2/C3/C4/C5)
-> biodynamic_day_type(element -> root/leaf/flower/fruit)
-> vastu_zone_prescriptions(lat, lon, radius)
-> guild_relations(primary_plant)
-> local_species_filter(pfaf, ecoregion)
-> planting_calendar(30-day window)
```

But the pipeline does not end at the planting calendar. It extends into a living exchange frontend for seeds, cuttings, plant starts, and produce — timed to the cosmological calendar, located by vastu zone and ecoregion, with guild compatibility data attached to every listing. A farmer who plants Tulsi under Shravana nakshatra on a Jupiter day lists their surplus cuttings through the same system that told them when to plant. The buyer receives not just a cutting but the full cosmological context of its cultivation: the field state at germination, the guild companions it was grown alongside, the astrobotanical class and dosha profile of the harvest. This is what a produce market looks like when it is built on a knowledge graph rather than a logistics platform. The provenance is not a marketing claim. It is an immutable record in the attestation chain.

5.3 Federation Protocol and Node Transport

Each node — a van, a farm, a hackerspace, a gurukula, a research station, a traditional knowledge community — runs its own local-first Coherence Atlas instance. Shared canonical ontology (the stable shastra layer) provides interoperability across all nodes without imposing uniformity. Each node owns its sadhu

overlay layer entirely: what it contributes to the federation, at what authority level, and with what access permissions is determined by the node, not by any central authority.

The transport layer between nodes is pull-on-use: when a Florida node needs species data and cultivation observations that its local datasets do not hold, it pulls from a Kerala node whose practitioners have documented those plants across decades of nakshatra-grounded observation. That data arrives with its source node, lineage, and authority level intact — it is Kerala’s sadhu-layer knowledge, marked as such, enriching the Florida node’s planting calendar with specificity that no generic database provides.

Resilience is the natural consequence of genuine exchange rather than a designed backup strategy. Data that has been pulled and used is cached locally. If the Kerala node goes offline — flood, power failure, infrastructure loss — the knowledge it contributed to the federation survives in every node that pulled from it. The archive is distributed because the exchange was real. The community in Kerala that built that knowledge over generations retains full attribution in every node that holds it. Their contribution is immutable in the attestation chain regardless of what happens to their physical infrastructure.

Three principles govern the federation: canonical portability (the stable ontology moves between nodes without data loss or authority degradation), sadhu sovereignty (each node owns its overlay layer and controls its sharing), and guru transparency (AI inference is always marked as such and never promoted to canon without explicit confirmation from a node with shastra-level authority). No node can unilaterally alter another node’s canonical layer. Proposals from guru-layer synthesis are broadcast as proposals and confirmed or rejected by the receiving node’s own authority process.

Atlas-the-project is therefore not a service. It is an open-source codebase — the kernel, the engines, the coherence computation, the sound substrate, the Chladni yantra computation, the GSS epistemology — plus a reference instance running on kanjira in Gainesville. Each lineage, school, farm, or individual practitioner who runs Atlas is running their own instance with their own seed data and their own coherence calibration. The reference instance is one voice in the federation, not its center. This is not a scalability pattern. It is the structural guarantee that the software will never represent a lineage it has not been authorized to represent.

5.4 Coherence Spectrum

Atlas is the technical substrate for three creative projects sharing the same ontology. Coherence Spectrum is a persistent procedural civilization simulator in which Vedic frameworks function as gameplay mechanics; the coherence spectrum runs from 0% Kali Band to 100% Satya Band; dosha-based morphogenesis generates creature forms. Emergent Satya Yuga is the worldbuilding framework for the Kali-to-Satya transition; NPUs as Sanskrit-grammar-based meaning-computation units. The Devi Field is the speculative eco-mythic novel — Anika Rao, Eli Mercer, Maya Alvarez — in which stabilization arrives through entrainment rather than apocalypse. The game simulates. The novel narrates. Atlas measures. Coherence is relational density. The Satya Yuga is the state where the graph is maximally coherent.

6 The Invitation: The Bhupura

The Bhupura is the outer square of the Sri Yantra, the bounding enclosure with gates in the four cardinal directions and guardians at the subcardinals. It is the layer at which the sacred diagram meets the world that will enter it. In a populated Bhupura, each gate admits a specific kind of practitioner arriving from a specific direction with specific offerings for specific ritual work inside. The gates are not redundant. They are the same protocol repeated through different cosmological orientations, and the repetition is what makes the Bhupura a square rather than a circle.

The invitation that follows is structured along the same pattern. Eight community gates. Each gate is addressed to practitioners arriving from a specific domain of practice with specific things to offer and specific things to receive in return. The reader who finds the section structurally repetitive is reading correctly — the repetition is the point, and each gate is a protocol.

The unique offering Coherence Atlas makes to each community below is the same in essence and different in application: bring us your dataset — your founding dates, your site locations, your clinical outcomes, your field observations, your manuscript corpus, your species records — and the system will show you what the cosmological field was doing in relation to your most significant events, whether your sacred sites cluster near pre-existing field nodes across traditions, whether your founding moments share cosmological signatures you did not consciously plan, what your data looks like from inside the full relational fabric of the Vedic knowledge system. This is the cross-community query capability that no single-domain tool can provide. Every community below is being invited into a relational field that makes their own knowledge more legible to themselves.

6.1 East Gate (Indra, Fire): Computational Jyotish

The nakshatra in Coherence Atlas is not a chart datum. It is a live field coordinate connected simultaneously to plant, sound, deity, marma, vastu zone, raga, passage corpus, and the current muhurta score for twenty-four categories of action. A jyotishi asking “what plants are indicated for someone in Shani mahadasha with Shravana lagna” receives a live graph traversal across domains no jyotish software currently touches. The dasha context modulates the plant recommendation, which modulates the raga, which modulates the therapeutic direction —

the chart becomes a key into the full relational field rather than a standalone prediction instrument.

Astrocartography extended with the land quality layer becomes something genuinely new: not just where your Jupiter line runs but what the sacred geography, ecoregion, nakshatra-plant profile, and vastu orientation of that line's terrain actually are. A person asking where to relocate for healing receives not planetary line coordinates but a full relational description of the landscape quality along those lines. The cross-community query also opens statistical research the tradition has always implied: natal chart data correlated against field states at key life dates, asking whether certain nakshatra transits cluster around reported experiences in ways the clinical literature could validate.

What Atlas needs: rigorous cross-validation of panchanga computations, dasha calculation testing, ayanamsa edge cases. Ask: run /field against your preferred platform for thirty dates and document discrepancies. If your research involves historical chart correlation, bring the dataset — we will build the query interface.

6.2 North Gate (Kubera, Water): Sanskrit and Digital Humanities

A new Sanskrit annotation in the Coherence Atlas corpus does not merely add text. It opens a new school of interpretation that propagates through the entire graph simultaneously. A Shaiva reading of a cosmological entity becomes a traversable path that surfaces different plants, different ragas, different muhurtas for a practitioner aligned with that school. The overlay architecture preserves the interpretive plurality that the tradition has always held without collapsing it into a single canonical answer — Samaya and Kaula, Shaiva and Vaishnava, regional and pan-traditional readings coexist as separate authority layers on the same canonical entities, each fully queryable, none overwriting the others.

Research replication is architecturally guaranteed: the passage corpus is versioned and attestation-tracked. A scholarly finding can be reproduced against the same corpus state that generated it, with the full provenance chain visible. Cross-community query means a Sanskrit scholar's annotation on a specific Devi's dhyana propagates immediately into the sound layer (which raga correspondence does this reading suggest?), the plant layer (which nakshatra plant resonates with this interpretation?), and the geometric layer (does this reading alter the yantra form?). The graph is a live research environment.

What Atlas needs: corpus expansion — Ayurvedic granthas not yet in machine-readable form, Gandharva Veda, Grihya Sutras, the full Gaudiya corpus beyond

Narottama, Shaiva Agamas, Shakta Tantras. NLP tools for Sanskrit compound analysis. Federation protocol design with GRETIL and Vedavaapi so that scholarly annotation infrastructure can interoperate with the living system. Engagement with Gerard Huet’s Sanskrit Heritage work, Rick Briggs’s foundational observations on Sanskrit as a natural AI language, and the broader Paninian NLP literature. Ask: review the passage corpus schema. Identify priority ingestion sources. If you have a corpus already in machine-readable form, bring it.

6.3 South Gate (Yama, Earth): Biodynamic and Permaculture

The Thun sidereal calendar maps planting types to zodiacal constellations. Coherence Atlas maps them to nakshatras, adds the graha lord, element, guna, tithi quality, and devi resonance, scores every plant in the dataset against the live field state, and returns guild companions alongside the primary plant recommendation. This is the same practice carried further into its own relational logic — not a replacement for what you already do but a deepening of the cosmological grounding beneath it.

The cross-community query for farmers specifically includes the research layer: the DOK trial has 21 years of yield data. Coherence Atlas has the retroactive panchanga for every day of those 21 years. Running those yield records against the field state is a research question nobody has asked because nobody had a system that held both datasets simultaneously. Do high-yield years cluster around specific nakshatra transits? Do preparation applications made under certain field conditions outperform others? The system can answer these questions with existing data. Bring the dataset.

The exchange frontend extends this into commerce: a farm growing by nakshatra calendar lists seeds, cuttings, plant starts, and produce with full cultivation provenance attached — field state at germination, planting muhurta, biodynamic preparation used, guild companions, astrobotanical class, dosha profile. Buyers are not purchasing a product. They are purchasing a specific moment of the field’s expression.

What Atlas needs: field validation of nakshatra-plant correspondences, local species datasets, biodynamic preparation timing records, guild plant performance observations, and growers willing to pilot the exchange frontend. Ask: run /bhumu/agriculture for one planting season. Keep field notes. Return them as sadhu-layer annotations.

6.4 West Gate (Varuna, Water): Gaudiya Vaishnava Communities

This system was built from within the sampradaya. Every architectural decision derives from Gaudiya Vaishnava siddhanta taken seriously as a technical specification. What the system offers back is something that has not existed before: a computational instrument through which the tradition’s own internal coherence becomes visible and navigable in ways that deepen rather than flatten the knowledge.

The cross-community research capability is specifically significant here. The correlation of Prabhupada’s 108 founding communities against the sacred geography layer — mound complexes, Shakti Pithas, sacred water confluences, astrocartography lines from his natal chart — is a research question the system can answer that no other tool can approach. The panchanga at each founding date, the deity installed, the naming correspondence with Vraja geography, the land qualities of each location run against the bhumi vastu body map: this is a genuine research contribution about Prabhupada’s own work made visible through cosmological computation. Sanskrit context layering means that every new pada contributed to the corpus opens new traversal paths through the full knowledge graph — new connections between the poetry, the deity, the nakshatra, the plant, the sound, that the tradition has always known implicitly and that the graph makes explicit and researchable.

What Atlas needs: theological review of the knowledge frame, pada corpus expansion beyond Narottama, validation of the Vraja forest mappings, field observation from practitioners who can identify when cosmological scoring is resonant versus discordant. Ask: read the Goloka documentation. Tell us what the tradition requires that the system is not yet providing.

6.5 Northeast Gate (Ishana, Ether): Sri Vidya and Tantric Lineages

The Nitya Devi registry holds all fifteen Devis with weapons, mudras, vahana, hue, garments, ornaments, and capability signature vectors, with Samaya and Kaula interpretations preserved as separate overlay layers on the same canonical entities. A new Kaula reading of Kulasundari’s weapons does not just add an annotation — it propagates through the sound layer, the geometric layer, the plant layer, and the temporal layer. Cross-community query means a Sri Vidya practitioner can ask what the biodynamic community has observed about plants associated with specific Devis, or what the sound healers have found clinically about ragas that share a capability signature with a specific Devi’s weapons.

If your lineage holds knowledge that should not be publicly distributed, the node sovereignty model allows contribution to a restricted overlay layer accessible only to initiated practitioners — the infrastructure is already built for exactly this use case.

What Atlas needs: practitioners with genuine lineage standing to populate and validate the Devi dataset, contribute missing vahana and directional attributes, review capability signature vectors, and contribute Kaula overlay datasets. Ask: review the Devi dataset schema. Propose corrections or extensions.

6.6 Northwest Gate (Vayu, Air): Sound Healers and Music Therapists

The clinical evidence base for raga therapy is real: peer-reviewed controlled trials documenting Bhimpalasi for BP and HRV modulation, Todi for blood pressure and respiratory rate reduction across a 30-day intervention, Miyan ki Todi and Malkauns for attention modulation measured by ERP. The Pitta Raga Module provides a fully specified clinical protocol verified against Ayurvedic theory and clinical outcomes. What does not yet exist is a computational tool that integrates this evidence with the live cosmological field.

The Coherence Atlas system returns not just a raga but the full relational context of that raga at this moment: the nakshatra it derives from, the graha that governs it, the dosha it modulates, the time window in which it is most potent, the formant frequencies of the associated bija mantra running as a continuous drone beneath it, the marma points that resonate with it in the body. A sound healer working with this system is not selecting from a list of evidence-based ragas. They are entering a live field that holds the clinical evidence, the cosmological timing, the somatic map, and the specific quality of the present moment simultaneously.

Research replication is architecturally supported: every raga recommendation is derivable from the field state that generated it. A clinical practitioner can log session outcomes against the field state at the time of each session and ask whether certain nakshatra-tithi-vara combinations correlate with better outcomes for specific therapeutic applications.

What Atlas needs: clinical practitioners willing to run the system alongside their existing practice and document where the cosmological recommendations align with or diverge from clinical judgment. Ask: contact us with your clinical context. We will build a therapeutic interface adequate to your practice.

6.7 Southwest Gate (Nirriti, Earth): Indigenous and Traditional Ecological Knowledge

The federation architecture was designed with your situation specifically in mind. A community running its own Coherence Atlas node owns that node entirely. Nothing leaves without explicit decision to share it. The cryptographic attestation chain means that if knowledge is shared, it is permanently attributed to its source and cannot be extracted without the attribution chain following it.

The cross-community query works in both directions without compromising sovereignty: your knowledge of a specific plant can be correlated against the Ayurvedic layer without leaving your node. The sacred geography layer already holds 28 North American mound complexes and is actively expanding. Your community's site knowledge, contributed at whatever level of detail you determine appropriate, enriches the correlation analyses that benefit every node in the federation. If your physical infrastructure is threatened, the knowledge your community contributed remains in every node that pulled from it, permanently attributed to you.

What Atlas needs: traditional ecological knowledge communities willing to pilot the node sovereignty model and help define what genuine data sovereignty looks like in practice. Ask: reach out. We will build the node infrastructure to your specifications before asking for anything in return.

6.8 Southeast Gate (Agni, Fire): Hackerspaces, Makers, and Open Hardware

The attestation architecture in Coherence Atlas is a general-purpose knowledge provenance system that happens to be populated with traditional Vedic knowledge. The schema — subject | predicate | object | source | confidence | stability | authority | attestation — applies to any domain where the integrity of information matters and where attribution chains need to survive forks and derivatives. A maker community that builds on the Coherence Atlas federation protocol inherits immutable part compatibility data, BOM integrity verification, replacement part query substitution against a relational graph, and schematic versioning with provenance, using the same attestation infrastructure that the Vedic knowledge layer runs on.

This is not a side use case separate from the main project. It is the same structural problem meeting the same structural solution at a different layer of manifestation: how knowledge with custodians and authority can be shared

across a distributed network without its integrity being lost in transmission. The Bhupura's southeast gate belongs to Agni — action, transformation, practice — and the makers who enter through it are welcomed as practitioners of the same principles applied to a different material.

What Atlas needs: hardware community engagement with the attestation schema, pilot deployments of the federation protocol for parts inventory and BOM provenance, contributions to the node transport layer, and the documentation and distribution infrastructure that open hardware communities have always built well. Ask: read the schema. Propose extensions. Build your own node.

6.9 How to Connect

The system runs at localhost:5000 on kanjira in Gainesville, Florida. The /field endpoint requires no authentication and will tell you exactly what cosmological moment you are reading this in.

The offering cycle is the onboarding protocol:

impetus	-> what brought you here
cosmic context	-> what field state are you arriving in
offering plate	-> what you are bringing
seed	-> how it connects to what already exists
research lattice	-> what questions it opens
construction	-> software, wiki, dataset, visualization
git push	-> committing to the record
next cycle	-> the impetus that emerges

If you are reading this document, you have already completed the first step.

7 Appendix A: Current System State

As of April 2026, the morning of Dvādaśī śukla in Uttara Bhādrapadā.

Working: 217 routes in `kernel.py`. 27 engines, zero import failures. 163 Python files across 15 subdirectories in `npu_engine/`. 110+ CSV files across 33 knowledge domains. 7,208 entities, 8,466 relational edges. 26,000-chunk JSONL passage corpus with vector search via local Ollama embeddings. `/field` returning complete panchanga plus sound plus plant plus devi plus marma plus dasha plus natal on every request. Sound field loop live, tracking the field state every sixty seconds, tanpura at $Sa=196\text{Hz}$ (G3) for Uttara Bhādrapadā's ether element. Agriculture PWA (offline-capable). Talachakra Studio (tabla synthesis, MIDI output, panchanga integration). Vishvakarma (GIS vastu, satellite imagery, 9 zones). Lila Streams (land design, permaculture mandala, sacred sites). Bandhu figure (6 rendering layers, toroidal breath animation, doctrine of signatures). Ring engine (10 ring types). 219 wiki pages. All six domain routes returning 200. Chladni yantra interference engine live in the browser at `/s4` with interactive tithi, k, and alpha sliders. Fifteen tithi symmetry snapshots verified (C_3 , C_6 , C_{11} quasicrystalline, C_{15}). GPU benchmark: 1024×1024 Chladni wave field at 1.18ms on Intel Arc iGPU ($2.1 \times$ faster than CPU). Lunisolar gear ratio finding confirmed: $365.25/29.53 \approx 12.368$ produces maximum Sri Yantra alignment; Ekadashi at $k=12.0$ reaches 85% alignment, Purnima at $k=12.368$ reaches 82% full bloom. Research paper draft at `docs/atlas_paper_v02.typ`.

Specified, not yet built: Chladni yantra engine as Python module in `npu_engine/` (currently client-side JavaScript only). Real-time audio sync between sound engine and Chladni parameters. GPU-accelerated live Chladni rendering in the kernel. Sound Layer 2 (foreground-yield relational mixing graph, 11 layers specified and ready). Sound Layer 3 (probabilistic walk over augury graph). Per-user natal layer with computed natal tithi as first-class field. Shabda-kosha word-entity table with phonetically-computed gunas, doshas, and graha signatures. Full archana ingestion pipeline starting with Lalita-sahasranama. SuperCollider to MOTU audio output unblocked on PipeWire 1.2.6 JACK shim. Vocal SynthDef (1,154 lines, no audio path). 4D projection (code exists, not exposed). Deha app beyond Bandhu placeholder. Seeds pipeline (4 briefs, not yet feeding graph). Journal UI (backend exists, no HTML). Editor (specced, not built). Systemd services (scripts ready, not installed).

Dataset gaps: 30+ major Ayurvedic granthas. Grihya Sutra ritual calendar. Carnatic melakarta-to-nakshatra mapping. Vraja forest coordinates. Vahana and directional data for 3 Nitya Devis. Yoni cross-species dataset. The complete fifteen-Nitya archana data from Achala's *Journey with the Moon*.

8 Appendix B: Core Schemas

Entity: entity_id | entity_type | name | name_key | bija | theta | phi | s_layer
| element | guna | dosha | stability | authority

Relation: subject | predicate | object | source | confidence | stability |
authority | attestation

Passage: chunk_id | source_text | tradition | entity_refs | domain | authority
| passage_text

Herb: herb_id | name_sanskrit | name_latin | ay_rasa | ay_virya |
ay_vipaka | ay_dosha_effect | tcm_nature | tcm_flavor | west_primary_action
| biodynamic_day_type | nakshatra_correspondence | astrobotanical_class |
convergence_score | attestation

Attestation levels: OBSERVED:PRIMARY_TEXT (1.0) | OBSERVED:TRADITIONAL (0.8)
| OBSERVED:PEER_REVIEWED (0.9) | OBSERVED:PFAF (0.7) | SYNTHESIS (0.4) |
INTERPRETATION (0.2)

9 Appendix C: The Offering Cycle

impetus	-- what is drawing attention right now
cosmic context	-- what field state is this arising in
offering plate	-- what tasks and research directions are proposed
seed	-- how this connects to existing graph structure
research lattice	-- what questions this opens
construction	-- software, wiki pages, datasets, visualizations
README expansion	-- documenting what was built
git push	-- committing to the record
next cycle	-- the impetus that emerges from the work

10 Appendix D: Glossary

Acintya bheda abheda — inconceivable simultaneous difference and non-difference; the philosophical foundation of the Atlas graph architecture, from Sri Chaitanya Mahaprabhu's siddhanta.

Archana — structured ritual form in which a deity is worshipped by reciting a nāmāvalī (garland of names) with specific offerings at each name; functions as the tradition's own per-deity relational database in paddhati texts, and as the data-structural model Atlas inherits for per-seat deity content.

Ashtakala — eight eternal periods of Goloka lila; the S0 temporal mapping layer.

Chladni descent — the structural principle that cosmological meaning compresses as spatial frequency rises across the toroidal interference field; the principle by which this document is organized.

Dakshina marga — right-hand path; agamic, structured approach; corresponds to the shastra authority layer.

Dasha — Vimshottari planetary period system; biographical temporal context.

Lila — divine play; the ontological status of the system.

Lunisolar gear ratio — 365.25 solar days divided by 29.53 synodic lunar month days, approximately 12.368; the spatial frequency at which the toroidal interference field produces maximum alignment with the classical Sri Yantra; the physical basis of the tradition's designation of Ekadashi as high-vibration.

Muhurta — auspicious time window; scored for 24 action types by `intention_engine.py`.

Nakshatra — one of 27 sidereal lunar mansions; primary field coordinate and coherence-scoring variable.

Nama-rupa — name and form; the two axes of every graph entity. In the deeper shabda-brahman reading, the word is ontologically prior to its participations, and the participations (deity, plant, person) inherit the word's own computable qualities.

Panchanga — five-limbed almanac (tithi, nakshatra, vara, yoga, karana); the computational root of `field_state()`.

Phason — in quasicrystal physics, the moveable degree of freedom in a non-crystallographic symmetry pattern; in this document, the structural name for

Part III, the moment the paper steps outside its own cosmological descent to survey the software landscape.

Sampradaya — lineage of transmission; encoded as tradition weight in the authority overlay system.

Sat-sandarbha — Jiva Gosvami’s six philosophical treatises establishing Gaudiya Vaishnava epistemology; the structural model for Atlas’s knowledge architecture.

Seva — selfless service as offering; the exchange model of the federation.

Shabda — sound, word, name as a real cosmological entity prior to its referents; the ontological ground on which the Atlas entity model rests.

Shruti — microtonal interval; each carrying a named semantic quality from the Sangitaratnakara.

Tithi — lunar day (Sun-Moon angular separation divided by 12°); maps to Nitya Devi, modulates rasa.

Vama marga — left-hand path; Kaula, transgressive approach; corresponds to the sadhu overlay layer.

Yantra — geometric compression of cosmological relations; in the Chladni descent model, the natural attractor of the toroidal interference field at a specific spatial frequency.

Yoni — animal archetype per nakshatra; cross-species relational mode signature.

11 Appendix E: Primary Sources and Lineages

Gaudiya Vaishnava: Bhagavad Gita As It Is (A.C. Bhaktivedanta Swami Prabhupada trans. and commentary), Srimad Bhagavatam (Bhagavata Purana), Brahmasamhita, Chaitanya-charitamrita. Narottama das Thakura: Prarthana, Premabhakti-candrika (primary Bandhu corpus). Srila Rupa Gosvami: Bhakti-rasamrita-sindhu. Srila Jiva Gosvami: Sat-sandarbha. Srila Bhaktivinoda Thakura: Tattva-viveka, Jaiva Dharma. Srila Vishvanatha Chakravarti Thakura.

Sri Vidya and Nitya Devi tradition: Nityasodashikarnava; Yoginihrdaya; Lalita-sahasranama with Bhaskararaya’s Saubhagya-bhaskara commentary; Parashurama-kalpa-sutra. Modern consolidated reference: Achala Sylwia Mihajlović, *Journey with the Moon: Tithis in Spirituality and Astrology* (2018, 329 pages, discoveringyouniverse.com), fifteen per-Nitya chapters in canonical Sri Vidya sequence.

Jyotish: Brihat Parashara Hora Shastra, Jaimini Upadesha Sutras, Brihat Samhita (Varahamihira), Brihat Jataka (Varahamihira), Saravali (Kalyana Varma), Surya Siddhanta. Computational: Swiss Ephemeris with JPL DE431.

Cosmology: Rigveda (10.121, 1.25, 10.14), Atharvaveda (10.7 Skambha hymn), Vishnu Purana (H.H. Wilson trans.), Bhagavata Purana (3.11.18-19), Brahmanda Purana, Matsya Purana (ch. 273).

Music: Natya Shastra (Bharata, Manomohan Ghosh trans.), Sangitaratnakara (Sarngadeva, Kunhan Raja trans.), Brihaddeshi (Matanga). Clinical: Nagarajan-Varma Pitta Raga Module; Bhimpalasi BP/HRV RCT; Todi 30-day intervention; Miyan ki Todi / Malkauns / Puriya ERP study.

Ayurveda: Charaka Samhita, Sushruta Samhita, Bhavaprakasha Nighantu, Ash-tanga Hridayam. Open datasets: Amidha (CC BY 4.0, 700+ herbs), PFAF (8,504 species), ETCM (403 herbs).

Biodynamic: Steiner, R. Agriculture Course (GA 327). DOK trial (21-year controlled comparison, Switzerland).

Nakshatra traditions: Taittiriya Samhita (Keith trans.); ICAR Navagraha Vatika and Nakshatra Vatika.

Sanskrit grammarian tradition: Panini, Ashtadhyayi; Bhartrhari, Vakyapadiya (the shabda-brahman foundation of the Atlas entity model). Digital humanities: Gerard Huet, Sanskrit Heritage site; Rick Briggs, "Knowledge Representation in Sanskrit and Artificial Intelligence" (NASA AI Magazine, 1985).

Tantra and yantra computation: Kulaichev and others on Sri Yantra geometric construction methods; the Coherence Atlas Chladni descent paper (in preparation) on the lunisolar gear ratio finding.

Ritual: Shrauta Sutras, Grihya Sutras, Shatapatha Brahmana.